09b — ACID Properties

Key Concepts

Property	One-Line Definition
Atomicity	All or nothing — transaction fully succeeds or fully rolls back
Consistency	DB always moves from one valid state to another — constraints never broken
Isolation	Concurrent transactions don't interfere or see each other's uncommitted changes
Durability	Committed data survives crashes — persisted to disk, not just memory

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How It Works

- **Atomicity:** If any step in a transaction fails → entire transaction rolls back
- **Consistency:** Foreign keys, unique constraints, rules always enforced
- **Isolation:** Transactions behave as if they ran sequentially, even when concurrent
- **Durability:** Data written to disk (via WAL) before confirming commit

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Isolation Levels (Low → High Protection)

Level	What It Allows	Problem It Causes
Read Uncommitted	Read dirty (uncommitted) data	Dirty reads
Read Committed	Only read committed data	Non-repeatable reads
Repeatable Read	Same read returns same value always	Phantom reads
Serializable	Full isolation — transactions run sequentially	Lowest concurrency, slowest

> Most databases default to **Read Committed**

Real-World Scenarios

Scenario	ACID Property
Payment fails halfway	Atomicity rolls it back
Duplicate email signup blocked	Consistency enforces unique constraint
Two users book last seat simultaneously	Isolation ensures only one succeeds
Server crash after payment confirmation	Durability persists the committed data

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Industry Examples

- **Banks / PayPal** — every money movement is atomic; no partial transfers
- **Amazon** — order placement (charge + inventory + shipment) wrapped in one transaction
- **Airline systems** — isolation prevents double-booking the same seat

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Interview Questions

Question	One-Line Answer
What does ACID stand for?	Atomicity, Consistency, Isolation, Durability
What is atomicity?	Transaction fully succeeds or fully rolls back — no partial updates
Consistency vs Isolation?	Consistency = data rules enforced; Isolation = concurrent transactions don't interfere
Why is durability important?	Committed data survives crashes — written to disk, not just memory
What if DB is not ACID compliant?	Risk of partial writes, data corruption, lost updates, dirty reads
Name the 4 isolation levels?	Read Uncommitted, Read Committed, Repeatable Read, Serializable